1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 analysis shows co-existing but divergent microenvironmental signals in TNBC — B-cell/TLS features (CD20, TIL/TLS) that usually predict better outcomes coexist with macrophage/inflammatory signatures (CHI3L3) and vascular phenotypes (CD34). The puzzle is: why do lymphoid structures/TILs sometimes fail to translate into improved survival in TNBC, and what microenvironmental mediator(s) determine when immune organization is functionally beneficial? 2) Tier 1 paths used: I built primarily on CORE PATH 3 (CD20/TIL/TLS axis) and CORE PATH 1 (CHI3L3/macrophage-inflammation). I prioritized CD20 because TLS and B-cell biology are tightly connected to the chosen imaging feature (TLS-like lymphoid aggregates), and CHI3L3 because it provides a plausible macrophage-driven immunosuppressive mechanism that could negate TLS benefit. I deliberately deprioritized CD34 from the core hypothesis to keep the mechanistic question focused on immune organization vs. immune suppression. 3) Tier 2 support: GE profiles and multigene assays (supporting path 2 and 4) provide transcriptional classifiers to identify TLS-rich and CHI3L3-high tumors; CCND1/GLI1/Hedgehog and MKI67 (supporting path 1 and Tier2 entities) offer orthogonal axes (proliferation/stemness) that may modulate immune efficacy; baseline hemoglobin offers a systemic modifier affecting outcomes and treatment tolerance. These elements will be treated as secondary modifiers to refine stratification and test mechanism. 4) Endpoint choice: I chose overall survival (OS) because TCGA-BRCA provides robust OS data for hypothesis testing and because the central puzzle concerns why organized immune infiltration does or does not translate to durable clinical benefit. 5) Exclusions: I excluded Tier 3 exploratory entities (visceral adiposity index, isoprostanes, HR-negativity as a general fact) from the hypothesis to avoid diluting the mechanistic focus; CD34 (vascularity) was left out of the core claim to respect the Focusing Rule (max 2 anchors) though it remains relevant as a secondary axis for future work?

Working Hypothesis

Primary hypothesis: In TNBC, TLS-like lymphoid aggregates are prognostically favorable only when intratumoral macrophages exhibit a low CHI3L3-driven inflammatory program; high CHI3L3 activity remodels TLS niches into an immunosuppressive microenvironment that blunts B-cell–mediated antitumor immunity, thus negating TLS-associated survival benefits. Consequently, TLS-rich tumors stratified as CD20-positive/CHI3L3-low will have superior overall survival (OS) compared with TLS-rich/CD20-positive/CHI3L3-high tumors. A subset of TLS-rich/CHI3L3-high tumors will show activity that may be unmasked by macrophage-targeted therapies (e.g., CHI3L1/CHI3L3 blockade or CSF1R inhibition), suggesting a targeted combination approach to realize TLS-derived benefits. Secondary modifiers (CD34 vascularity, CCND1, GLI1, SHH, MKI67, and baseline hemoglobin) will modulate the magnitude of TLS benefit and identify contexts in which macrophage-directed strategies are most effective.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	CD20, CHI3L3
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CD20: CD20 because TLS and B-cell biology are tightly connected to the chosen imaging feature (TLS-like lymphoid aggregates), and CHI3L3 because it provides a plausible macrophage-driven immunosuppressive mechanism that could negate TLS benefit.

CHI3L3: CHI3L3-high tumors; CCND1/GLI1/Hedgehog and MKI67 (supporting path 1 and Tier2 entities) offer orthogonal axes (proliferation/stemness) that may modulate immune efficacy; baseline hemoglobin offers a systemic modifier affecting outcomes and treatment tolerance.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CD34	Supporting modifier from knowledge graph
Secondary	CCND1	Supporting modifier from knowledge graph
Secondary	GLI1	Supporting modifier from knowledge graph
Secondary	SHH	Supporting modifier from knowledge graph
Secondary	MKI67	Supporting modifier from knowledge graph
Secondary	Baseline hemoglobin	Supporting modifier from knowledge graph
Secondary	GE_profiles / multigene assays	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: the imaging anchor (TLS-like aggregates linked to CD20),

Prediction 2: the macrophage/inflammatory axis (CHI3L3), and

Prediction 3: secondary axes (proliferation/stemness signals via CCND1/GLI1/SHH/MKI67 and systemic modifiers like baseline hemoglobin) as modulators of TLS efficacy.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CD34: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CCND1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.